CAWM-Mamba

A Unified Model for Infrared-Visible Image Fusion and Compound Adverse Weather Restoration

Presented by: Sajjan Singh

ASIP Lab
IISER Bhopal, India

Presentation Roadmap

Part I: Background

- Introduction to Image Fusion
- The Adverse Weather Challenge
- Limitations of Current Methods
- Our Key Contributions

Part II: Methodology

- Overall Architecture Design
- Weather-Aware Preprocess (WAPM)
- Cross-modal Interaction (CFIM)
- Wavelet State Space Block (WSSB)

Part III: Experiments

- Experimental Setup & Datasets
- Quantitative & Qualitative Results
- Downstream Task Performance
- Computational Efficiency

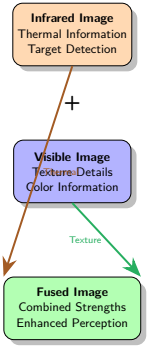
Part IV: Critical Analysis

- Identified Limitations
- **Proposed Solutions**
- Future Research Directions
- Conclusions

Introduction to Multimodal Image Fusion

What is Multimodal Image Fusion (MMIF)?

- Integrates **complementary information** from various imaging modalities
- Produces clearer, more informative fused images
- Critical for autonomous driving, UAV monitoring, military reconnaissance



Infrared (IR) Images:

- Penetrates smoke and obstructions
- Highlights thermal targets effectively
- **Limitation:** Low resolution, lacks fine texture

Visible (VI) Images:

- High-resolution texture and color details
- **Limitation:** Susceptible to adverse weather
- Suffers from light attenuation and distortion

The Adverse Weather Challenge

Critical Real-World Applications:

- **Autonomous Driving:** Real-time perception in rain, haze, snow conditions
- **UAV Monitoring:** Aerial surveillance under varying weather
- **Military Reconnaissance:** Target detection in degraded environments

The Core Problem

Existing IVIF methods are developed under ideal conditions and primarily focus on interference-free scenarios. They often **fail to provide reliable fusion outcomes** in complex weather scenarios!

Impact of Different Weather Types:

Haze	Rain	Snow
Contrast loss	Directional streaks	Particle occlusion
Color shifts	High-frequency	Multi-directional
Low-frequency	Local interference	Texture masking

Limitations of Current Methods

Challenge 1: Single-Degradation Focus

- Current methods handle **only one type** of degradation
- Fail when **multiple degradations coexist**
- Examples: Haze&Rain, Rain&Snow, Haze&Snow

Challenge 2: Two-Stage Pipeline

- Rely on manual preprocessing
- Restoration $\rightarrow$ Fusion (sequential)
- Synchronization issues
- Loss of scene details

Challenge 3: Frequency Overlap

- Rain/Snow: High-frequency components
- Haze: Low-frequency components
- Different processing strategies needed
- **No unified mechanism exists**

Our Solution:

Unified end-to-end framework that handles **compound degradations** with **single-pass** inference!

Our Key Contributions

Contribution 1: Novel Framework

CAWM-Mamba: A unified end-to-end framework for image fusion with compound weather restoration

- **Unified shared weights**
- Multiple degradations simultaneously
- Robust fusion in complex environments
- Single-pass inference

Contribution 2: WAPM & CFIM

Weather-Aware Preprocess Module:

- Embeds global weather information
- Enhances visible features

Cross-modal Feature Interaction:

- Effective feature alignment
- Multi-scale pooling

Contribution 3: WSSB

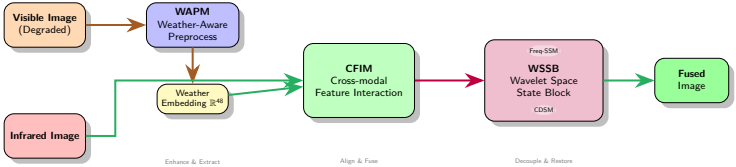
Wavelet Space State Block:

- Wavelet-domain decomposition
- Decouples multi-frequency degradations
- **Freq-SSM:** Anisotropic scanning
- **CDSM:** Unified degradation space

Achieved Results

- **AWMM-100K:** Consistent SOTA
- **Standard datasets:** Superior performance
- **Downstream:** Enhanced detection & segmentation
- **Efficiency:** Only 371G FLOPs

Overall Architecture of CAWM-Mamba



WAPM
Enhance degraded features
Extract weather embeddings
Global context awareness

CFIM
Cross-modal alignment
Multi-scale pooling
Feature complementation

WSSB
Wavelet decomposition
Freq-SSM scanning
Unified degradation space

Weather-Aware Preprocess Module (WAPM)

Design Motivation:

- Weather conditions **significantly affect** visible light image quality
- Need to extract weather-related information before fusion
- Provide explicit prior guidance for subsequent modules

Three-Stage Processing:

1. **Feature Extraction:**
   - Two convolutional layers with ReLU
   - Captures degradation-related information
2. **Channel Attention:**
   - Global Average Pooling (GAP)
   - Sigmoid activation for channel weights
3. **Global Weather Embedding:**
   - Vector $F_{embedding} \in \mathbb{R}^{B\times 48}$
   - Contains global weather information

Key Equations

$$\text{atten} = \sigma(W_2(\sigma(W_1(\text{GAP}(V)))))$$

$$V_{enhance} = \text{atten} \odot V_{feat}$$

$$F_{embed} = W_4(\sigma(W_3(\text{GAP}(V_e))))$$

Benefits

- Enhances weather-related features
- Provides global weather context
- Guides subsequent restoration
- Lightweight computation

Cross-modal Feature Interaction Module (CFIM)

Core Challenge: How to effectively fuse IR and VI features under compound weather interference?

Stage 1: Feature Modulation

IR Feature Processing:

- F_{ir}^{mp}: MaxPooling $\rightarrow$ Salient features, thermal contours
- F_{ir}^{gmp}: Global MaxPooling $\rightarrow$ Scene statistics

VI Feature Processing:

- F_{vi}^{ap}: AvgPooling $\rightarrow$ Texture features
- F_{vi}^{gap}: Global AvgPooling $\rightarrow$ Global statistics

Stage 2: Cross-modal Fusion

Attention Weight Generation:

$$A_{vi}^{ap} = \text{Softmax}(\sigma(\text{Conv}(P_{ir}^{mp})))$$

Cross-modal Enhancement:

- IR saliency $\rightarrow$ enhances VI texture
- VI details $\rightarrow$ optimize IR boundaries

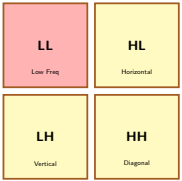
Output: $F_{out} = \text{Concat}[F_{ir}^{out}, F_{vi}^{out}]$

Key Insight: Simple element-wise summation fails to capture the complex non-linear correlations between modalities.

Wavelet Space State Block: Frequency Analysis

Why Wavelet Transform?

- **Simultaneous time-frequency localization**
- Pinpoint **spatially variant degradations**
- **Directional separation**: H, V, D subbands



DWT Decomposition

Frequency Characteristics

Low-frequency (LL):

- Haze degradation
- Global structure
- Brightness information

High-frequency (LH,HL,HH):

- Rain streaks
- Snowflakes
- Directional details

Key Insight

Rain/Snow → High-freq
Haze → Low-freq

Different scanning strategies needed!

Freq-SSM: Frequency-Selective State Space Model

Directional Alignment

Freq-SSM aligns scanning with subband direction:

- **LH subband:** Vertical scanning
- **HL subband:** Horizontal scanning
- **HH subband:** Diagonal scanning

Benefits:

- Captures dependencies along principal axis
- Ignores irrelevant orthogonal directions
- Maximizes restoration efficiency
- **No computational redundancy**

Selective State Space Model

Discretized SSM Equations:

$$\bar{A} = \exp(\Delta A)$$
$$\bar{B} = (\Delta A)^{-1}(e^{\Delta A} - I) \cdot \Delta B$$

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = Ch_t + Dx_t$$

Key Advantage

- **Linear complexity** $O(n)$
- vs. Transformer's $O(n^2)$
- Global effective receptive field
- Input-dependent projections

Freq-SSM: Algorithm Pipeline

WSSB Forward Process

Require: Feature map X_{in}
Ensure: Restored Feature X_{out}

$\{X_{LL}, X_{LH}, X_{HL}, X_{HH}\} \leftarrow \mathrm{DWT}(X_{in})$
for each $X_{freq} \in \{X_{LH}, X_{HL}, X_{HH}\}$ **do**
  $X_{norm} \leftarrow \mathrm{LayerNorm}(X_{freq})$
  $X_{scan} \leftarrow \mathrm{DirectionalScan}(X_{norm})$
  $X_{ssm} \leftarrow \mathrm{SSM}(X_{scan})$
  $X_{freq}^{out} \leftarrow X_{freq} + \mathrm{Linear}(X_{ssm})$
end for
$X_{LL}^{out} \leftarrow \mathrm{RegularSSM}(X_{LL})$
$X_{out} \leftarrow \mathrm{IDWT}(\{X_{LL}^{out}, \ldots\})$
return X_{out}

Key Operations

1. **Wavelet Decomposition:**
   - Haar wavelet, $J = 1$ level
   - Zero-padding for boundaries
2. **Directional Scanning:**
   - Align with degradation direction
   - Reduce redundancy
3. **Residual Connection:**
   - Preserve original information
   - Learnable scaling factor
4. **Channel Attention:**
   - Select key channels
   - Avoid channel redundancy

Common Degradation Space Mechanism (CDSM)

Design Purpose: Improve robustness against varying composite degradations

Mechanism Implementation

Lightweight Residual Block:

$$A_i = \text{Conv}(\sigma(\text{Conv}(A_{i-1}))) + A_{i-1}$$

Channel Attention:

$$\text{att} = \sigma(\text{Conv}_1(\sigma(\text{Conv}_1(\text{AP}(A)))))$$

Global Residual:

- Adds processed features to input
- Preserves multi-level information

Weather-Guided Modulation

Integration with WAPM:

- Weather embeddings guide processing
- Three-stage post-processing gate

Gating Process:

1. Extract global spatial statistics (GAP)
2. Concatenate with weather embedding
3. MLP: Linear $\rightarrow$ ReLU $\rightarrow$ Linear $\rightarrow$ Sigmoid
4. Generate channel weights $w \in [0, 1]^C$

Output: $X_{out} = X_{refined} \odot w$

Result: Dynamic frequency-selection gate that filters residual degradation noise!

Comprehensive Loss Functions

Wavelet Loss

Frequency-domain consistency:

$$L_{wavelet} = L_{low} + \frac{L_{LH} + L_{HL} + L_{HH}}{3}$$

where $L_{ii} = \|I_{ii} - I_{ii}^{fused}\|_1$

Pixel-Level Losses

Intensity:

$$L_{int} = \frac{1}{HW}\|I_{fused} - \max(I_{vis}, I_{ir})\|_1$$

Color:

$$L_{color} = \frac{1}{HW}\|T_{CbCr}(I_{fused}) - T_{CbCr}(I_{vis})\|_1$$

Perceptual Losses

Perceptual:

$$L_{per} = \frac{1}{3}\sum_{l}\|\phi_l(I_{fused}) - \phi_l(I_{vis})\|_1$$

where ϕ_l = MobileNetV3 features

Gradient:

$$L_{grad} = \frac{1}{HW}\|\nabla I_{fused} - \max(\nabla I_{vis}, \nabla I_{ir})\|_1$$

Total Loss

$L_{Total} = \alpha L_{wavelet} + L_{color} + L_{per} + L_{grad} + L_{int} + L_{SSIM}$

α: **learnable scalar** (init. 0.5)

Experimental Setup

Training Configuration

- **Hardware:** 2× Nvidia RTX 3090
- **Optimizer:** Adam
- **Learning Rate:** 0.0001
- **Crop Size:** 64 → 144 (progressive)
- **Architecture:** U-Net style
- **Block Config:** [8,10,10,12,10,10,8]
- **Channels:** 48 → 384 → 48

Dataset: AWMM-100K

Total: 100,000 image pairs

Single Degradation:

- Rain/Haze/Snow: 700 train / 50 test each

Compound Degradation:

- H&R / R&S / H&S: 1000 train / 50 test each

✔ **Zero scene overlap**

Standard Datasets

LLVIP, MSRS, M3FD for clean fusion

Evaluation Metrics

Nine Metrics for Comprehensive Assessment:

Metric	Description	Optimal
Q_{MI}	Normalized Mutual Information	↑ Higher
Q_{NICE}	Nonlinear Correlation Metric	↑ Higher
Q_G	Gradient-Based Metric	↑ Higher
Q_M	Multiscale Scheme Metric	↑ Higher
Q_P	Phase Congruency Metric	↑ Higher
Q_S	Chen-Varsheny Metric	↑ Higher
Q_{CV}	Quality Assessment Metric	↓ **Lower**
SSIM	Structural Similarity	↑ Higher
Q_{abf}	Quality Assessment-Based	↑ Higher

Compared Methods

LRRNet, TarDal, SHIP, FusionBooster, MaeFuse, GIFNet, SAGE, FreeFusion, AWFusion

Note

Two-stage baselines rely on AdaIR/AwaCLe for restoration. AWFusion and CAWM-Mamba work end to end.

Quantitative Results: Compound Weather Scenarios

Performance on Haze&Rain, Rain&Snow, Haze&Snow:

Scene	Method	Q_{MI}	Q_G	Q_M	Q_P	Q_S	SSIM	Q_{abf}	Rank
Haze&Rain	AWFusion	0.275	0.351	0.409	0.216	0.775	0.337	0.410	3.00
	CAWM-Mamba	**0.360**	**0.376**	**0.743**	**0.240**	**0.844**	**0.381**	**0.509**	**1.00**
Rain&Snow	AWFusion	0.413	0.387	0.355	0.253	0.734	0.348	0.431	3.89
	CAWM-Mamba	**0.477**	**0.383**	**0.799**	**0.249**	**0.846**	**0.391**	**0.514**	**1.22**
Haze&Snow	AWFusion	0.286	0.343	0.383	0.195	0.759	0.338	0.399	3.11
	CAWM-Mamba	**0.368**	**0.377**	**0.712**	**0.236**	**0.840**	**0.392**	**0.511**	**1.00**

Key Observations

- **Near-perfect rank** (1.00-1.22) across all compound scenarios
- **Ranks first in ALL 9 metrics** under Haze&Rain and Haze&Snow
- Significant margin over AWFusion (3.00-3.89)
- Consistent superiority across all degradation combinations

Quantitative Results: Single Weather Scenarios

Performance on Rain, Haze, Snow:

Scene	Method	Q_{MI}	Q_G	Q_M	Q_P	Q_S	Q_{abf}	Rank
Rain	AWFusion	0.351	0.333	0.643	0.196	0.834	0.462	3.78
	CAWM-Mamba	**0.431**	**0.390**	**0.758**	**0.236**	**0.845**	**0.502**	**1.00**
Haze	AWFusion	0.264	0.428	0.547	0.359	0.774	0.532	3.56
	CAWM-Mamba	**0.501**	**0.539**	**0.882**	**0.486**	**0.837**	**0.634**	**1.00**
Snow	AWFusion	0.467	0.355	0.554	0.272	0.812	0.494	3.00
	CAWM-Mamba	**0.532**	**0.433**	**0.746**	**0.349**	**0.837**	**0.566**	**1.00**

Outstanding Achievement

- **Perfect average rank of 1.00** across ALL three single weather conditions
- Advantage even more pronounced than compound scenarios
- Especially strong in **Haze scenario** with large margins across all metrics
- Visual evidence confirms superior removal of directional artifacts (rain streaks, snow)

Visual Comparison: Fusion Quality Analysis

Haze&Rain Scene

Two-stage methods (e.g., FusionBooster):

- Fail to disentangle interference
- Result in blurring and "halo" artifacts

AWFusion (end-to-end baseline):

- Noticeably darker background
- Lower contrast
- Insufficient atmospheric correction

CAWM-Mamba Results

- **Optimal brightness restoration**
- **Sharp structural edges**
- **Natural color fidelity**
- Eliminates directional interferences
- Retains background texture

Key Evidence:

- Road markings preserved
- No residual rain streaks
- Clear target boundaries

Insight: WAPM's global weather embedding guides optimal restoration, while Freq-SSM ensures directional artifact removal.

Downstream Task Performance

Object Detection (YOLOv7)

Method	mAP@0.5	mAP@0.5:0.95
VI Only	0.571	0.339
IR Only	0.347	0.203
AWFusion	0.781	0.488
CAWM-Mamba	**0.826**	**0.528**

Per-category gains:

- Person: +0.022
- Car: +0.023
- Bus: +0.031

Semantic Segmentation (BANet)

Method	mIoU (%)
VI Only	59.76
IR Only	39.55
AWFusion	68.38
CAWM-Mamba	**75.98**

Highest IoU in ALL categories!

- Car: 86.93%
- Person: 69.24%
- Bike: 68.91%

Statistical Significance

A one-sided paired t-test gives $p = 0.032 < 0.05$, confirming that the gains are statistically significant.

Computational Efficiency Analysis

Model Complexity Comparison:

Method	Params	FLOPs
AWMFuse	137.29M	1145.5G
Text-IF	89.01M	1518.88G
TG-ECNet	160.86M	—
AWFusion	19.36M	4167.20G
CAWM-Mamba	**31.95M**	**371.01G**

Efficiency Highlights

- **Lowest FLOPs:** Only 371.01G
- **$\sim$1/11 of AWFusion** (4167.20G)
- Suitable for **UAV deployment**
- Latency-sensitive applications

Why So Efficient?

- **Mamba:** Linear $O(n)$ complexity
- **vs. Transformer:** Quadratic $O(n^2)$
- Optimal size-speed trade-off
- Real-time inference possible

Identified Limitations of CAWM-Mamba

⚠ Limitation 1: Extreme Concurrent Degradations

Problem: Performance challenges with ≥ 3 simultaneous degradations

Examples:

- Haze & Rain & Snow combined
- Multiple overlapping artifacts

Root Cause:

- **Frequency Masking Effect**
- Rain, snow, noise overlap in high-freq bands
- Spectral congestion limits Freq-SSM
- May cause minor residual noise

⚠ Limitation 2: Single-Frame Focus

Problem: Current evaluation focuses on single-frame processing

Unexplored Scenarios:

- Ground-level autonomous driving videos
- Dashcam sequences
- Continuous video streams

Root Cause:

- Temporal consistency not leveraged
- Potential flickering in video
- Dynamic environment instability

→ **Next:** We propose comprehensive solutions for each limitation!

Solution for Limitation 1: Multi-Level Wavelet & Adaptive Routing

Problem: Frequency overlap when $\geq$3 degradations coexist

Solution 1A: Multi-Level DWT

Approach:

- Extend to J=2,3 levels
- Finer frequency separation
- Scale-specific processing

Implementation:

- Level 1: Coarse separation
- Level 2: Fine-grained
- Hierarchical Freq-SSM
- Cross-scale attention

Benefit:
Different degradations can be isolated at different scales!

Solution 1B: Adaptive Router

Approach:

- Learn to route degradation types
- Specialized branches
- Soft routing via attention

Architecture:

- Degradation classifier
- Type-specific modules
- Dynamic fusion
- Mixture of Experts

Benefit:
Optimal processing path for each degradation combination!

Solution 1C: Spectral Unmixing

Approach:

- Blind source separation
- Wavelet domain unmixing
- Endmember extraction

Inspiration:

- Hyperspectral unmixing
- Independent Component Analysis
- Non-negative matrix factorization

Benefit:
Separate overlapping spectral signatures!

Combined Strategy: Multi-level DWT + adaptive routing + spectral unmixing for robust triple-degradation handling.

Solution for Limitation 2: Video Extension Strategies

Problem: Single-frame processing lacks temporal consistency

Solution 2A: Temporal Mamba

Core Idea:

- Extend 2D-SSM to **3D-SSM**
- Add temporal dimension
- Maintain $O(n)$ complexity

Architecture:

- Spatial SSM (per frame)
- Temporal SSM (across)
- Spatiotemporal fusion

Complexity:
Still $O(n \cdot t)$ where t = frames

Benefit:
Native video understanding!

Solution 2B: Flow-Guided

Core Idea:

- Optical flow for motion compensation
- Propagate across frames
- Ensure temporal coherence

Pipeline:

- Estimate optical flow
- Warp adjacent features
- Attention aggregation
- Bidirectional propagation

Benefit:
Reduces flickering artifacts!

Solution 2C: Recurrent Memory

Core Idea:

- Hidden state across sequence
- Memory mechanism
- Long-range dependencies

Implementation:

- $h_t = \text{GRU}(x_t, h_{t-1})$
- ConvLSTM integration
- Memory attention

Benefit:
Efficient video processing with memory!

Recommendation: Temporal Mamba offers the best efficiency–accuracy trade-off for real-time video processing.

Ablation Studies: Component Analysis

Impact of Each Component

WAPM Removal:

- Significant drop in Haze scenarios
- Confirms weather embedding importance
- Q_{MI}: 0.360 → 0.323

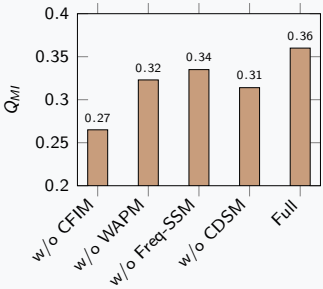
Freq-SSM Removal:

- Notable drop in Rain/Snow scenarios
- Proves anisotropic scanning value
- Q_{MI}: 0.360 → 0.335

CFIM Removal:

- Decline across all metrics
- Simple summation insufficient
- Q_{MI}: 0.360 → 0.265 (largest impact!)

Quantitative Impact



Key Insight

Modules are effective **individually** and **synergis-**

Conclusion: Key Takeaways

Key Contributions

1. **First unified framework** for joint fusion and compound weather restoration
2. **Three novel modules:**
   - WAPM: Weather-aware preprocessing
   - CFIM: Cross-modal interaction
   - WSSB: Wavelet space state block
3. **State-of-the-art** on AWMM-100K and standard datasets
4. **Superior downstream** performance
5. **Efficient:** Only 371G FLOPs

Real-World Impact

Autonomous Driving:

- Robust perception in adverse weather
- Enhanced safety margins

UAV Monitoring:

- All-weather surveillance
- Efficient edge deployment

Military & Security:

- Target detection in degraded environments
- Multi-sensor integration

Take-Home Message:

CAWM-Mamba demonstrates that **unified end-to-end processing** of compound degradations with multi-modal fusion achieves superior results with remarkable efficiency. The key innovations are **frequency-aware processing** via Freq-SSM and **cross-modal alignment** via CFIM.

Future Research Directions

Short-Term Goals

1. **Extended Benchmark:**
   - Triple-degradation dataset
   - Video fusion benchmark
   - Real-world dashcam sequences
2. **Architecture Improvements:**
   - Multi-level wavelet
   - Temporal SSM integration
   - Efficient inference
3. **Downstream Integration:**
   - End-to-end perception
   - Task-specific fine-tuning

Long-Term Vision

1. **Multi-Modal Extension:**
   - LiDAR + Camera fusion
   - Radar integration
   - Event camera support
2. **Foundation Model:**
   - Large-scale pretraining
   - Zero-shot generalization
   - Universal restoration
3. **Real-World Deployment:**
   - Edge device optimization
   - Real-time processing
   - Safety-critical applications

Thank You!

Questions & Discussion

github.com/Feecuin/CAWM-Mamba

arXiv preprint: 2603.02560

sajjan25@iiserb.ac.in